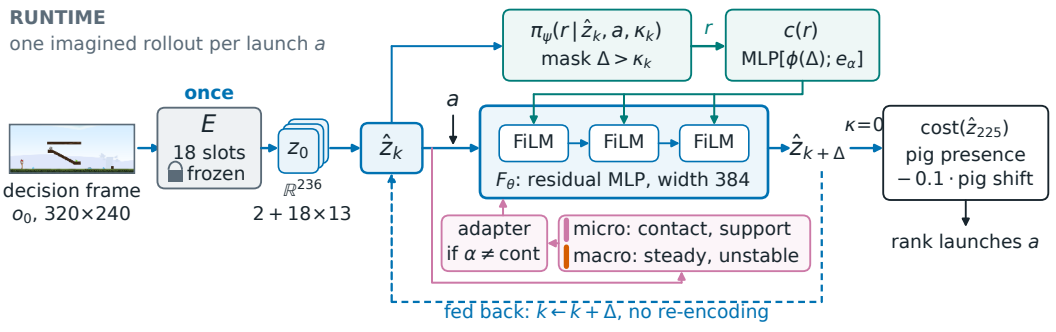

1 Encoder E
engine-projected slot labels; trained once, then frozen

2 Predictor F_θ · 480 fit levels
local: teacher-forced + recursive, 4 Δ-steps
long: recursive, horizon 60 → 225 frames
rank: success < failure cost (fit verdicts)
targets $E(o_t)$; predicates: engine labels
9 requests round-robin

3 Policy π_ψ · 120 policy levels
DP labels, fixed context:
$\Delta \cdot \mathrm{MSE} + \lambda \cdot$ compute ratio
→ cross-entropy distillation,
2 aggregation rounds
(J_{dec}: decision-valued labels)